

Find the voltage V_2 flowing through the resistor R_2 in Figure 1.

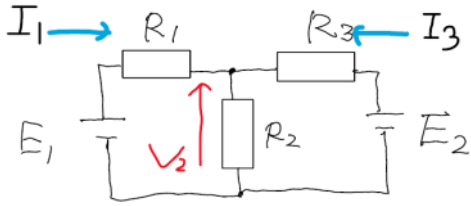
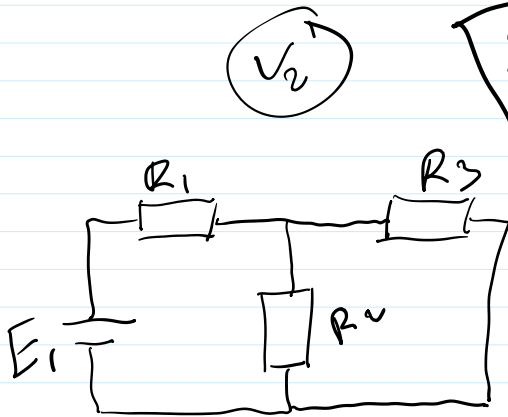
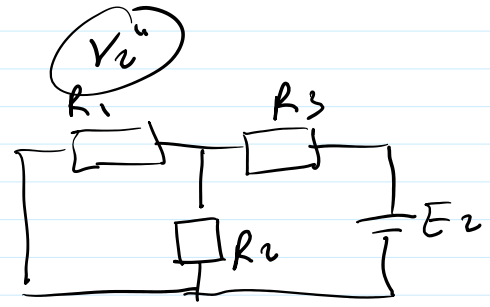
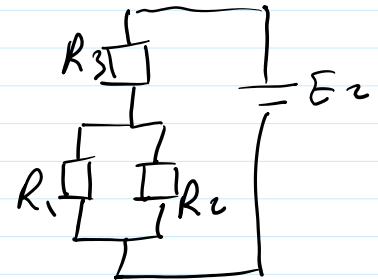


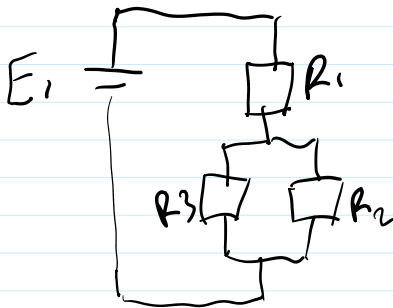
Fig 1



using the principle of superposition we find V_2 for each of the power supplies independently.



$$V_2'' = \frac{\frac{R_1 R_2}{R_1 + R_2}}{\frac{R_1 R_2}{R_1 + R_2} + R_3} E_2$$



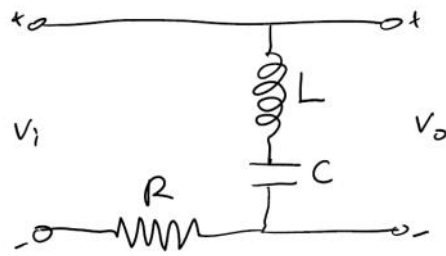
$$V_2' = \frac{\frac{R_2 R_3}{R_2 + R_3}}{\frac{R_2 R_3}{R_2 + R_3} + R_1} E_1$$

$$V_2 = V_2' + V_2''$$

$$V_2 = \frac{\frac{R_2 R_3}{R_2 + R_3}}{\frac{R_2 R_3}{R_2 + R_3} + R_1} E_1 + \frac{\frac{R_1 R_2}{R_1 + R_2}}{\frac{R_1 R_2}{R_1 + R_2} + R_3} E_2$$

$$V_2 = \dots = \frac{R_1 R_2 E_2 + R_2 R_3 E_1}{R_1 R_2 + R_2 R_3 + R_3 R_1}$$

Q2



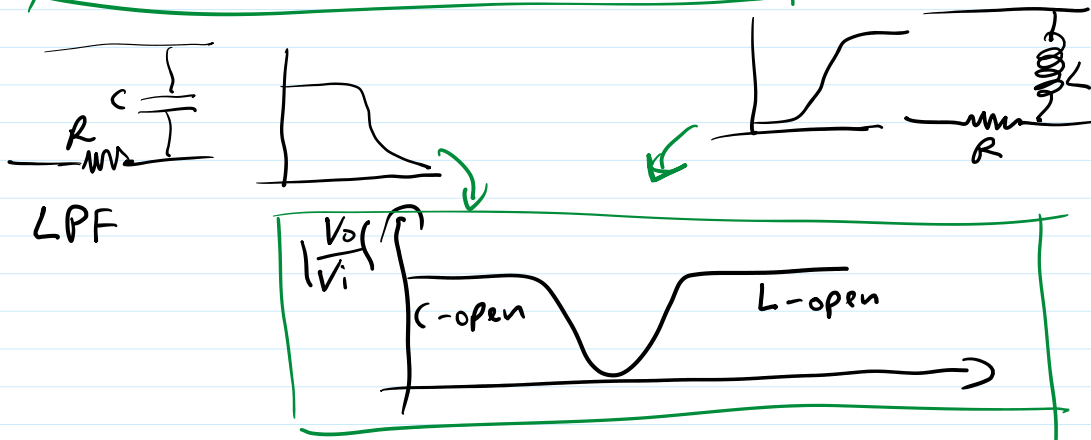
1. Find the transfer function of the filter circuit above
2. Draw the general shape of the frequency response of this filter
3. Describe the function of this filter

(1.)

$$\frac{V_o}{V_i} = \frac{Z_L + Z_C}{R + Z_L + Z_C}$$

$$= \frac{j\omega L + 1/j\omega C}{R + j\omega L + 1/j\omega C}$$

$$\left| \frac{V_o}{V_i} \right| = \left| \frac{j(\omega L - \frac{1}{\omega C})}{R + (j(\omega L - \frac{1}{\omega C}))} \right|$$



- ⊕ The capacitor lets low frequencies pass
- ⊕ the inductor lets High frequencies pass
- ⊕ only a certain frequency "band" in the middle will be blocked.

Therefore this is called a band-block filter
or band-stop filter.

Q3

- 1) Using the npn transistor with grounded Emitter shown in the figure below as an example, explain the principle of operation of the transistor.

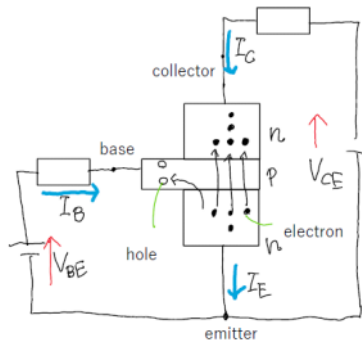


Figure 1: Emitter grounding of npn-type transistor

This transistor can work as a digital switch in the cut-off and saturation region or the static property, or as an amplifier in the active region depending on the base current and the load resistor.

The Base-Emitter is a p-n junction, when it is biased forward by V_{BE} it opens n-p-n path from the collector to the emitter $\Rightarrow$ Large current can run from C to E, depending on the value of V_{CE} & the load resistor.

- 2) Consider the grounded base configuration below instead, how will the operation of the operation differ from the circuit above.

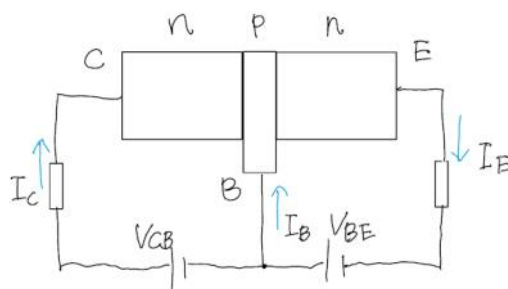


Figure 2: Base grounding of npn-type transistor

The common base config. will prevent the output voltage from feeding back to the input $\Rightarrow$ the transistor works as an isolator.

(*) forward bias of the Base-Emitter junction is determined by the difference between V_{BE} & V_{CE}

(*) input current is I_C , output current is I_E

$\Rightarrow$ current amplification factor $\beta = \frac{I_o}{I_i} = \frac{I_E}{I_C} \leq 1$

- 1) Find the voltage amplification factor of the circuit shown in Fig. 1
- 2) In the circuit shown in Fig. 1, find and show the output voltage when the signal shown in Fig. 2 is input.

$$R_B = 10 \text{ [k}\Omega\text{]}$$

$$A_c = 2 \text{ [kL]}$$

$$h_{FE} = 200$$

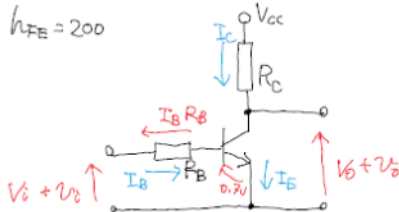


Fig. 1

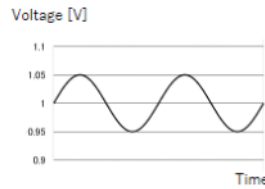


Fig. 2 Input voltage (1)

- under the conditions of 1) and show it in the figure.

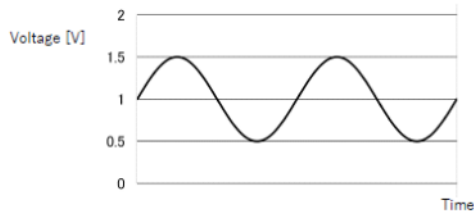


Fig. 3 Input voltage (2)

Q

$$A_v = \frac{\Delta v_o}{\Delta v_i} = - \frac{h_{fe} R_C}{R_B}$$

$$= - \frac{200 \times 2 \times 10^3}{10 \times 10^3}$$

$$Aa = -40$$

$$\textcircled{2} \quad V_o = V_{cc} - I_c R_c$$

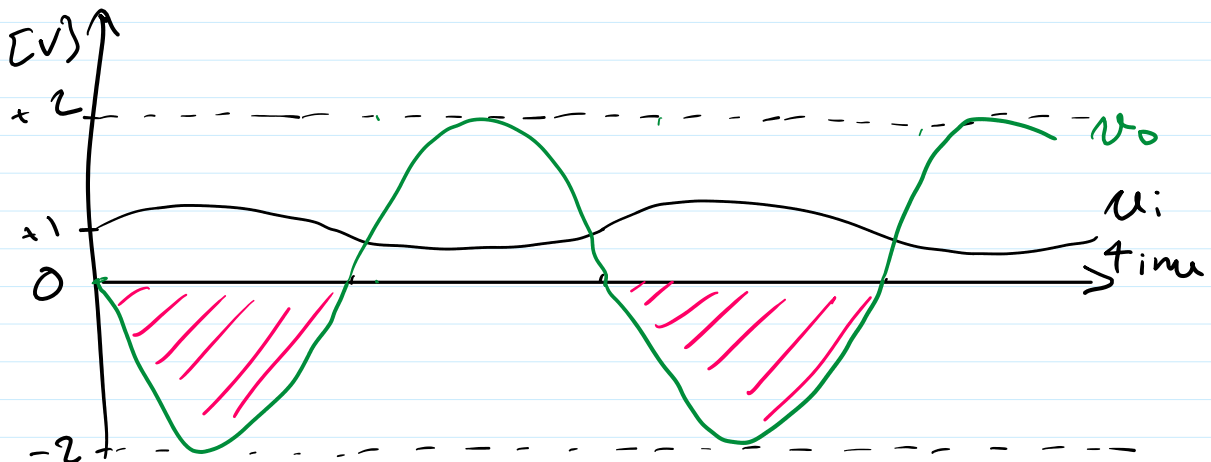
$$= 12 - \frac{\ln_2(V_i - 0.7)}{R_B} R_C$$

$$= 12 - \frac{200(-0.7)}{10 \times 10} \times 2 \times 10$$

$$V_o = 12 - 12 = 0V!!$$

U; peak-to-peak = 0.1 V

$$V_o \text{ peak to peak} = |-40 \times 0.1| = 4 \text{ V}$$

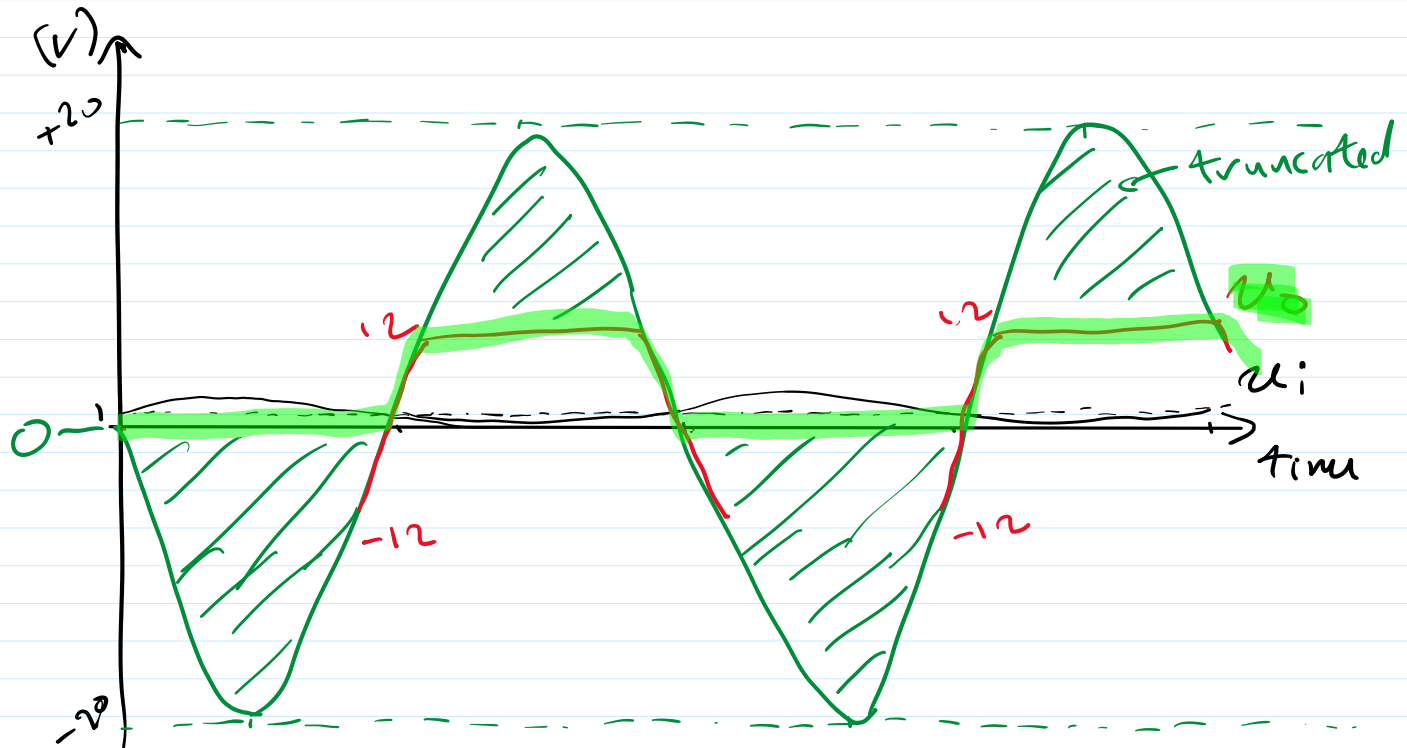


③

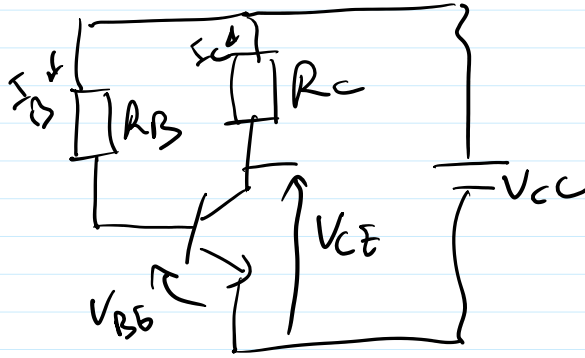
$$V_o = 0V$$

$$u_i \text{ peak to peak} = 1V$$

$$u_o \text{ peak to peak} = |-40 \times 1| = 40V$$



Q5:



① find I_C :

$$I_C = h_{fe} I_B = h_{fe} \frac{V_{CC} - V_{BE}}{R_B}$$

② find V_{CE} :

$$V_{CE} = V_{CC} - I_C R_C$$

$$= V_{CC} - h_{fe} \frac{V_{CC} - V_{BE}}{R_B} R_C$$

$$V_{CE} = \left(1 - \frac{R_C}{R_B} h_{fe}\right) V_{CC} + \frac{R_C}{R_B} h_{fe} V_{BE}$$

done...